



Xaurora: Advancing subseasonal-to-seasonal forecasting by fine-tuning foundation weather models with spectral consistency

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Sub-seasonal to seasonal (S2S) timescales range from two weeks to three months and are crucial to make informed climate change-related decisions, including renewable energy resources allocation, extreme events' risks mitigation, and the development of effective early warning systems. Unfortunately, traditional physics-based forecasting systems achieve poor skill on these lead times. Recently, deep learning (DL) has shown promising results in weather forecasting on timescales up to 10 days, reaching performance competitive with that of physical models. However, these DL approaches currently struggle on S2S timescales.

Following previous studies on neural solvers for partial differential equations and weather forecasting, we propose a fine-tuning framework aimed at improving the S2S prediction skill of foundation weather models. Our approach has two core components. First, we implicitly condition the latent space embeddings to retain the predictable signals at a given lead time using an additional regression head. Second, we design a novel frequency-domain decoder and loss function to ensure spectral consistency. These steps should ensure that the model focuses on the most predictable frequencies. We apply this methodology to the recently published Aurora foundation model and propose *Xaurora*, standing for "extended Aurora". Our fine-tuning approach represents an important milestone in data-driven S2S forecasting, addressing key challenges in the field while remaining broadly applicable with minimal assumptions on the underlying model's architecture.

The relevance of our framework is evaluated through ablation studies, comparing our spectral consistency fine-tuning to the original Aurora model. Furthermore, we provide standard deterministic and probabilistic skill scores on S2S timescales, as well as relevant teleconnection indexes. We present preliminary outputs of this analysis.